॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 11

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Docker Compose for Multi-Container App
Theory:	Docker Compose is a tool used to define and manage multi-container Docker applications using a single `docker-compose.yml` file. It automates the creation, networking, and management of multiple containers, enabling services such as web applications and databases to run together with a single command.

Implementation:	Step 1: Create a project folder named `my-multi-app`.
	`C:\Users\sharm\Desktop\7th sem\DevOpsL>mkdir my-multi-app`
	`C:\Users\sharm\Desktop\7th sem\DevOpsL>cd my-multi-app`
	Step 2: Create a `docker-compose.yml` file and define the web application and database services.

Welcome | docker-compose.yaml

docker-compose.yaml

```yaml
version: '3.8'

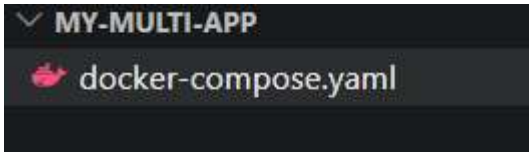
services:

  web:
    image: my-flask-app:v2
    ports:
      - "8080:5001"
    volumes:
      - my-app-data:/app/data
    depends_on:
      - redis

  redis:
    image: redis:latest
    ports:
      - "6379:6379"
    volumes:
      - redis-data:/data

volumes:
  my-app-data:
```

Step 3: Save the Compose file.



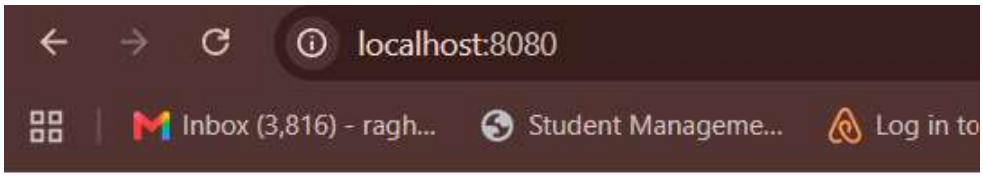
Step 4: Start all services using:
docker-compose up -d

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\my-multi-app>docker compose up -d
[+] up 3/3
 ✔Network my-multi-app_default   Created
 ✔Container my-multi-app-redis-1 Started
 ✔Container my-multi-app-web-1   Started
```

Step 5: Verify running containers using:
docker compose ps

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\my-multi-app>docker compose ps
NAME                  IMAGE              COMMAND                  SERVICE   CREATED          STATUS          PORTS
my-multi-app-redis-1  redis:latest       "docker-entrypoint.s…"   redis     33 seconds ago   Up 33 seconds   6379/tcp
my-multi-app-web-1    my-flask-app:v2    "python app.py"          web       33 seconds ago   Up 33 seconds   0.0.0.0:8080->5001/tcp, [::]:8080->5001/tcp
```

Step 6: Access the application through http://localhost:8080.



Hello, Docker!

2026-07-23 04:51:19: Hello, Docker!

Step 7: Verify communication between the web application and database service.

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\my-multi-app>docker network ls
NETWORK ID     NAME                   DRIVER    SCOPE
2be1d6e458df   bridge                 bridge    local
0bca7cc88be1   host                   host      local
8460cc27a411   my-multi-app_default   bridge    local
46d93f010b5f   none                   null      local
```

Step 8: Stop all services using:
docker-compose down

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\my-multi-app>docker-compose down
[+] down 3/3
 ✔Container my-multi-app-web-1   Removed
 ✔Container my-multi-app-redis-1 Removed
 ✔Network my-multi-app_default   Removed
```

Subject Faculty Name: Prof. Arijit Dutta